Discrete Mathematics I Test 1 - 12.03.2026

Solve the following problem: A logging system generates short diagnostic codes consisting of four symbols chosen from the set $\{A, B, C, D, E\}$.

The order of the symbols does not matter, but symbols may be repeated. For example, $AABC$ and $CAAB$ represent the same code.

How many different diagnostic codes are possible?

Solution

Version 1 - stars and bars

Since the order of symbols does not matter and repetition is allowed, the problem is equivalent to selecting a multiset of size 4 from the set $\{A, B, C, D, E\}$.

Let

$$x_A, x_B, x_C, x_D, x_E$$

denote the number of times each symbol appears in the code. Then the total number of symbols is 4, so

$$x_A + x_B + x_C + x_D + x_E = 4,$$

where

$$x_A, x_B, x_C, x_D, x_E \geq 0.$$

Thus we must count the number of nonnegative integer solutions of this equation.

Using the *stars and bars* method, the number of solutions is

$$\binom{4+5-1}{5-1} = \binom{8}{4}.$$

Therefore, the number of different diagnostic codes is

$$\binom{8}{4} = 70.$$

Hence, there are $\boxed{70}$ different diagnostic codes.

Version 2 - Direct counting

Another way to solve the problem is to classify the possible multiplicities of the symbols appearing in the code.

Since four symbols are chosen, the possible multiplicity patterns are:

$$4, \qquad 3+1, \qquad 2+2, \qquad 2+1+1, \qquad 1+1+1+1.$$

We count the number of codes corresponding to each pattern.

- **Case** 4**:** all four symbols are the same (e.g. $AAAA$). Choose the symbol:

$$\binom{5}{1} = 5.$$

- **Case** $3+1$**:** three symbols are the same and one is different (e.g. $AAAB$). Choose the symbol appearing three times and the different symbol:

$$5 \cdot 4 = 20.$$

- **Case** $2+2$**:** two symbols appear twice (e.g. $AABB$). Choose the two symbols:

$$\binom{5}{2} = 10.$$

- **Case** $2+1+1$**:** one symbol appears twice and two others appear once (e.g. $AABC$). Choose the repeated symbol and the two remaining symbols:

$$5 \cdot \binom{4}{2} = 30.$$

- **Case** $1+1+1+1$**:** all symbols are different (e.g. $ABCD$). Choose four symbols:

$$\binom{5}{4} = 5.$$

Adding all cases gives

$$5 + 20 + 10 + 30 + 5 = 70.$$

Thus, there are $\boxed{70}$ diagnostic codes.

Version 3 - Path counting

The equation

$$x_A + x_B + x_C + x_D + x_E = 4$$

describes the number of nonnegative integer solutions.

Using the stars and bars interpretation, we represent the four selected symbols by stars and separate the five symbol types by four bars. For example,

$$**\,|\,*\,||\,*\,|$$

corresponds to

$$x_A = 2,\ x_B = 1,\ x_C = 0,\ x_D = 1,\ x_E = 0.$$

Each solution corresponds to an arrangement of 4 stars and 4 bars. Equivalently, it corresponds to a lattice path from $(0,0)$ to $(4,4)$ consisting of

- 4 horizontal steps (stars),
- 4 vertical steps (bars).

The number of such paths is

$$\binom{8}{4}.$$

Therefore, the number of diagnostic codes equals

$$\binom{8}{4} = 70.$$

Version 4 - Multisets

Since the order of the symbols in the code does not matter, a diagnostic code can be viewed as a *multiset* of size 4 chosen from the set $\{A, B, C, D, E\}$.

Thus we are counting the number of multisets of size 4 whose elements come from a set of 5 symbols.

It is a standard result that the number of multisets of size k from a set of n elements equals

$$\binom{n+k-1}{k}.$$

In this problem we have $n = 5$ symbols and $k = 4$ positions, therefore

$$\binom{5+4-1}{4} = \binom{8}{4}.$$

Hence the number of possible diagnostic codes is

$$\binom{8}{4} = 70.$$

Version 5 - Generating function - will be later

Let x_A, x_B, x_C, x_D, x_E denote the number of occurrences of each symbol. We must count the nonnegative integer solutions of

$$x_A + x_B + x_C + x_D + x_E = 4.$$

For each symbol we may choose it $0, 1, 2, 3$, or 4 times. Thus the generating function for one symbol is

$$1 + x + x^2 + x^3 + x^4.$$

For five symbols the generating function becomes

$$(1 + x + x^2 + x^3 + x^4)^5.$$

The coefficient of x^4 counts the number of ways to obtain a total of four symbols.
Since the upper bound 4 is not restrictive here, we may replace

$$1 + x + x^2 + x^3 + x^4$$

by the geometric series

$$\frac{1}{1-x}.$$

Thus we obtain

$$\left(\frac{1}{1-x}\right)^5.$$

Using the binomial expansion

$$(1-x)^{-5} = \sum_{k=0}^{\infty} \binom{5+k-1}{k} x^k,$$

the coefficient of x^4 is

$$\binom{8}{4}.$$

Hence the number of codes is

$$\binom{8}{4} = 70.$$

Version 6 (Advanced math) - Burnside's Lemma.

Consider all ordered sequences of length 4 over the alphabet $\{A, B, C, D, E\}$. There are 5^4 such sequences.

Two sequences represent the same diagnostic code if one can be obtained from the other by permuting the four positions. Thus the symmetric group S_4 acts on the set of sequences.

We want the number of orbits of this action. By Burnside's Lemma,

$$\text{number of codes} = \frac{1}{|S_4|} \sum_{\sigma \in S_4} \mathrm{Fix}(\sigma),$$

where $\mathrm{Fix}(\sigma)$ is the number of sequences fixed by σ.
We compute this number for each cycle type.

- Identity (1)(2)(3)(4): every sequence is fixed.

$$\text{Fix} = 5^4.$$

- Transpositions (6 permutations): positions in the 2-cycle must match.

$$\text{Fix} = 5^3.$$

- Double transpositions (3 permutations): two pairs must match.

$$\text{Fix} = 5^2.$$

- 3-cycles (8 permutations): the three positions must match.

$$\text{Fix} = 5^2.$$

- 4-cycles (6 permutations): all four symbols must match.

$$\text{Fix} = 5.$$

Therefore

$$\frac{1}{24}\left(5^4 + 6 \cdot 5^3 + 3 \cdot 5^2 + 8 \cdot 5^2 + 6 \cdot 5\right) = \frac{1680}{24} = 70.$$

Thus the number of diagnostic codes is $\boxed{70}$.

Most common mistake

It is very common to see such a reasoning: "There are 4 places, 5 letters to use, so the number of possible sequences is $5^4 = 625$". So far so good, this is the maximal number of sequences. But some sequences are equivalent, e.g. AABC=ABAC=ABCA etc. So the argument goes: "So now I have to adjust to the equivalence of codes. There are 4 letters I might permute, so I will divide by 4!. So the answer is $5^4/4!$". But this is wrong reasoning and we should notice something is wrong as 5^4 is not divisible by $4! = 24$. So what is wrong?

Observe, that among all possible sequences (not equivalent codes), there is only one $AAAA$, there are also 4 equivalent codes: $AAAB$, $AABA$, $ABAA$, $BAAA$. We may even see the pattern:

1. $\frac{4!}{4!} = 1$ codes equivalent to $AAAA$
2. $\frac{4!}{3!} = 4$ codes equivalent to $AAAB$
3. $\frac{4!}{2!2!} = 6$ codes equivalent to $AABB$
4. $\frac{4!}{2!} = 12$ codes equivalent to $AABC$
5. $4! = 24$ codes equivalent to $ABCD$

This comes from the fact, that indeed we might perform 4! permutations, but some permutations do not change the sequence. But this also means, that the number of equivalent codes per class is not fixed globally, i.e. it is not always $4! = 24$, so we cannot divide by 4! to adjust for equivalency.

But in fact you might see that everything agrees when enumerating the codes directly. We've seen, that there are a 70 non-equivalent codes:

1. 5 codes of the form $AAAA$, for each there is only 1 equivalent code;

2. $5 \cdot 4 = 20$ codes of the form $AAAB$, for each there are 4 equivalent ones;

3. $\binom{5}{2} = 10$ codes of the form $AABB$, for each there are 6 equivalent ones;

4. $5\binom{4}{2} = 30$ codes of the form $AABC$, for each there are 12 equivalent ones;

5. 5 codes of the form $ABCD$, for each there are 24 equivalent ones;

If we sum those we obtain:

$$5 \cdot 1 + 20 \cdot 4 + 10 \cdot 6 + 30 \cdot 12 + 5 \cdot 24 = 5 + 80 + 60 + 360 + 120 = 625 = 5^4 \tag{1}$$

So we obtain all possible sequences splitted into equivalence classes.